A production-referenced evaluation of design-stage decision rules in deep drawing: when can a prediction replace a physical trial?

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Abstract

In recent years, simulation and machine learning have informed design-stage manufacturability decisions. However, they are judged by prediction error, not by whether their verdicts could replace a physical trial. This study offers a production-referenced evaluation of design-stage decision rules. Firstly, the within-run floor is introduced as its unit: the difference that drift and scatter of production and measurement produce between two batches of one design within a run. Then, for each evidence relation of a new design to produced ones (sibling, new setting or new family), a rule returning meets, fails or trial is characterized by its decisive distance, beyond which at least 95 % of its verdicts are correct, and its safe distance, beyond which at most 5 % are wrong. A guard band turns them into single verdicts. In an exploratory application to one laboratory deep-drawing process, with alternatives produced once and an uncalibrated simulation dataset, simulation-based, learned and production-record rules are scored. The nearest produced setting, one rule for every relation, decided at 3.4 floors with a sibling and 8.5 for a new setting. No simulation-based or learned rule was more decisive; a Gaussian process specified for small data reduced to a force trend. Borrowing the other family's errors on the same relation, not learning, made the production record safe for new settings. For a new family, only simulation-based rules decided, none within 30 floors. Within a family, a production record indexed by its relation to the coming design is the baseline learned manufacturability checks must beat.

Keywords: decision support, design for manufacturing, machine learning, uncertainty quantification, selective prediction, deep drawing

1 Introduction

Whether a design alternative can be made to its requirements is decided long before it is made. In sheet-metal forming, the decision is based on a finite-element (FE) simulation (Banabic, 2010): the design team compares the predicted characteristics with the requirements and then releases the alternative, changes it or sends it to a physical trial. In recent years, artificial intelligence (AI) has made this loop faster: surrogates replace simulations, models learn manufacturability from earlier and synthetic designs (Yan et al, 2026; Ghadai et al, 2018; Ballegeer and Benoit, 2026), and predictive-quality models and virtual metrology (VM) estimate the quality from process data (Tercan and Meisen, 2022; Dreyfus et al, 2022). However, such models are judged by their prediction error on held-out data. This leaves open the question that the team has to answer: *can this prediction be relied on to decide, or is a trial still needed?* Mechanical design cannot be verified by simulation in the way integrated circuits are (Whitney, 1996), and the answer depends on how much production has been seen and on how the new design relates to it.

Uncertainty quantification supplies intervals (Nemani et al, 2023), and VM attaches a reliance index to each prediction (Cheng et al, 2008). However, neither states how close to a requirement a verdict can be relied on, in a unit that production defines, for each relation of a new design to the produced evidence. Thus, three research questions (RQs) are addressed in this study. RQ1: How can the fitness of a design-stage prediction to decide be expressed on a production-referenced scale for any predictor and characteristic? RQ2: How does it depend on the amount of production evidence and on how the new design relates to it? RQ3: With six to nine produced alternatives, what do a generic simulation dataset, Gaussian processes (GPs), gradient boosting and calibrated intervals contribute compared with a production-record baseline?

In this study, a production-referenced evaluation framework for design-stage decision rules is proposed. Firstly, the *within-run floor* is introduced as its unit: the difference that drift and scatter of production and measurement produce between two batches of one design within a run. After that, a rule that turns the design-stage evidence into a verdict (*meets*, *fails* or a physical *trial*) is scored, for each *evidence relation* between a new design and the produced alternatives, by two points on its operating characteristic: its *decisive* and *safe distances*. Finally, a procedure turns these distances into a guard band for single verdicts and prices them by the costs of errors and trials. The decisions in scope are those of process planning, the approval of process changes in series production, where a customer may require a resubmission run, and the qualification of AI-based tools that support them, taken by a team that has already produced variants of a design family. A new family stands for the feasibility question before the tryout of a new die, whereas the tryout itself, whose tool changes between loops, is out of scope.

The framework is demonstrated on open data: 9,000 deep-drawn and cut cups of dual-phase steel, produced as 18 design–process alternatives of 500 consecutive parts each on one laboratory tool set (Baum and Heinzelmann, 2026), and FE simulations of the same process from a generic dataset (Baum and Heinzelmann, 2025). Each alternative was produced once, and the simulations were not calibrated to the process. Therefore, the study is exploratory: it shows whether the framework discriminates

between rules and evidence situations on real data, and its findings are hypotheses for confirmation. The main contributions of this study are the framework; an evaluation design that separates the evidence relations, isolates what the simulation adds and tests the learners against a production-record baseline; the findings for this process; and a proposed reporting checklist with an open scoring function.

The paper is organized as follows. Section 2 reviews the related work, Section 3 describes the framework, and Section 4 introduces the deep-drawing data and the evaluation design. Section 5 presents the results, Section 6 discusses them, and Section 7 concludes the paper.

2 Background and related work

2.1 Machine learning for manufacturability, predictive quality and virtual metrology

Design for manufacturing (DFM) codifies what a process can make into rules and cost models (Boothroyd et al, 2010). In recent years, machine learning (ML) has been used to learn manufacturability from computer-aided design (CAD) models (Ghadai et al, 2018; Zhang et al, 2018), synthetic sheet-metal datasets (Ballegeer and Benoit, 2026), and process data and simulated design spaces (Yan et al, 2026). Predictive-quality models and VM estimate the quality from process data (Wuest et al, 2016; Tercan and Meisen, 2022; Dreyfus et al, 2022), and their uncertainty is obtained by Bayesian, ensemble and conformal methods (Nemani et al, 2023; Heddoub et al, 2026). Automatic VM attaches to every prediction a reliance index and a similarity index that flags unfamiliar process data (Cheng et al, 2008, 2012); it is the closest analogue to the guard band of this study. Furthermore, models are carried across products and processes by transfer learning and domain adaptation (Maschler and Weyrich, 2021), also from simulation to real parts (Tercan et al, 2018). In sheet forming, such models predict defects and springback from simulated and measured data (Cao et al, 2024; Heinzelmann et al, 2025b), and the discrepancies between synthetic and real deep-drawing data are being addressed (Baum et al, 2026b). However, these models, as well as language models in case studies (Makatura et al, 2024) and benchmarks (Picard et al, 2025), are evaluated by their held-out accuracy or coverage, not by their decisions against production.

2.2 Simulation credibility and the validation of forming simulation

Verification and validation (V&V) ask whether a model is accurate enough for its intended use (Oberkampf and Roy, 2010), and credibility assessment scales the required evidence with the consequence of a wrong decision (ASME, 2018). Bayesian calibration estimates the calibration parameters together with a model discrepancy (Kennedy and O'Hagan, 2001; Higdon et al, 2004); without the discrepancy, the parameters are biased and over-confident (Brynjarsdóttir and O'Hagan, 2014; Arendt et al, 2012). Multi-fidelity models combine a cheap and an expensive model through a fitted scale and a discrepancy (Kennedy and O'Hagan, 2000; Le Gratiet and Garnier,

2014). In sheet forming, the main sources of simulation error are friction (Hol et al, 2012), the unloading and hardening behavior behind springback (Yoshida and Uemori, 2002; Wagoner et al, 2013) and tool deflection (Pilthammar et al, 2018). Robustness analysis propagates the input scatter (Wiebenga et al, 2014; Chen et al, 2022), and the draw-in is measured and controlled in line (Havinga et al, 2020; Fischer et al, 2018). In addition, simulation datasets support surrogate and transfer learning (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025a), and the deviation between simulation and part has been decomposed statistically (Baum et al, 2026a). However, how close to a requirement a calibrated or learned prediction can decide remains an open question.

2.3 Decisions with abstention and their evaluation

In conformity assessment, a measured value proves conformance, proves non-conformance or leaves the decision open. The guard bands are set by the measurement uncertainty (ISO, 2017), and the global consumer's and producer's risks follow from a prior distribution of the measurand (Joint Committee for Guides in Metrology, 2012). The operating characteristic (OC) curves of acceptance sampling and selection give the probability of a decision against the true quality, with an indifference zone inside which no decision is required (Bechhofer, 1954). Similarly, a classifier with a reject option trades errors against abstentions (Chow, 1970; Geifman and El-Yaniv, 2017), as does regression with a reject option (Zaoui et al, 2020), and learning to defer passes difficult cases to an expert (Madras et al, 2018). The errors are priced by cost-sensitive evaluation (Elkan, 2001) and decision curves (Vickers and Elkin, 2006). In addition, appropriate reliance on automation requires that its reliability be visible (Lee and See, 2004), although displayed confidence can calibrate trust without improving joint accuracy (Zhang et al, 2020, 2021). The decisive and safe distances of this study are such operating characteristics, and the guard band is the counterpart of a conformity-assessment one, which equally depends on a prior.

2.4 Production variation, measurement and design decisions

Statistical quality control compares the short- and long-term variation with tolerances through capability and performance indices (Montgomery, 2019). In an initial process study, an index above 1.67 meets the acceptance criteria, and an index between 1.33 and 1.67 may be acceptable after a review by the customer (Automotive Industry Action Group, 2006); time-dependent models describe drifting processes (ISO, 2026). Precision experiments define the repeatability limit, the difference that two results under repeatability conditions exceed with a probability of 5 % (ISO, 2025). They also define the intermediate precision under conditions that differ in time (ISO, 2023). Measurement systems analysis separates the repeatability from bias, linearity and stability (Automotive Industry Action Group, 2010). Design decisions between alternatives with uncertain outcomes (Hazelrigg, 1998) are taken with margins (Eckert et al, 2019) or with tests whose cost is traded against that of errors (Thomke and Bell, 2001; Salado and Kannan, 2018). Furthermore, what a team knows about a new alternative depends on whether it is a variant, an adaptation or an original design

Table 1 Positioning of the framework among the methods that support or evaluate design-stage manufacturability decisions

Approach	Evidence	Output	Scored against production	Abstains
DFM rules and cost models (Boothroyd et al, 2010)	codified process knowledge	guideline compliance, cost	no	no
V&V, credibility assessment (Oberkampf and Roy, 2010; ASME, 2018)	physics model, validation tests	accuracy; required evidence	on test cases	can require tests
Bayesian calibration (Kennedy and O'Hagan, 2001)	simulations, experiments	predictive distribution	not as decisions	implicitly
Conformal prediction (Lei et al, 2018)	exchangeable calibration data	interval with coverage	coverage	yes
Virtual metrology with reliance index (Cheng et al, 2008)	process data of produced items	value, reliance and similarity index	on held-out production	flags
Conformity assessment, OC curves (ISO, 2017; Bechhofer, 1954)	measurement and its uncertainty	accept, reject, undecided; risks	on the measured item	yes
Selective classification and regression (Chow, 1970; Zaoui et al, 2020)	labeled data	value or reject; risk–coverage	on held-out data	yes
ML manufacturability models (Ghadai et al, 2018; Yan et al, 2026)	historical or simulated designs	class or value	on held-out data	rarely
This study	simulations, produced alternatives	verdict; distances in floors by evidence relation	yes, per verdict	yes

(Pahl et al, 2007), and the evidence relations of this study mirror this distinction for production evidence.

Table 1 places the framework among these methods. The framework adopts the indifference zone of operating characteristics, the guard band of conformity assessment, the selective risk of abstaining predictors and grouped cross-validation (Roberts et al, 2017). The novelty of this study lies in the production-defined unit in which every rule and characteristic is scored, in the qualification by evidence relation, and in their combination into a procedure for design-stage decisions that any predictor can enter.

3 Production-referenced evaluation framework

In this section, the decision problem is defined first. After that, the within-run floor, the decision rules, their decisive and safe distances and the evidence relations are introduced. Finally, the procedure is given and illustrated by an example.

3.1 Decision problem and expected cost

A design alternative a is a combination of product geometry and process settings that is produced as a series of parts, and a quality characteristic c varies over these parts.

A requirement is an upper limit R on the characteristic, or on its absolute value for characteristics whose target is zero, that a share p of the parts must meet. Therefore, alternative a is *adequate* for c when the p-quantile of its parts satisfies $q_p(a, c) \leq R$. However, q_p is unknown at the design stage. A decision rule r maps the available evidence to an interval $[L_r, U_r]$ for q_p and to a verdict, which is given in Eq. (1).

$$V_r = \begin{cases} \text{meets} & U_r \leq R, \\ \text{fails} & L_r > R, \\ \text{trial} & \text{otherwise.} \end{cases} \tag{1}$$

The three verdicts mirror the decision rules of conformity assessment (ISO, 2017). A rule that returns a point ($L_r = U_r$) always decides. The evidence consists of the simulations of a and of the production record of other alternatives, the *calibration alternatives*. A verdict costs c_{FA} when a failing alternative is accepted, c_{FR} when an adequate one is rejected and c_{T} when it is sent to trial. Thus, the expected cost of a rule per verdict is

$$C = c_{\mathrm{FA}} P_{\mathrm{FA}} + c_{\mathrm{FR}} P_{\mathrm{FR}} + c_{\mathrm{T}} P_{\mathrm{T}}, \tag{2}$$

where P_{FA}, P_{FR} and P_{T} are the probabilities of a false accept, a false reject and a trial. The two distances of Section 3.4 summarize this cost at a fixed share of errors. Three limits of scope apply. Firstly, the demonstration uses upper limits on the 95th percentile ($p = 0.95$, for a normal characteristic a performance index of about 0.55), and lower limits follow by a change of sign. Secondly, the verdicts are per characteristic, and a design-level verdict over all characteristics is reported as an extension (Section 5.7). Thirdly, attribute outcomes such as splits and wrinkles, which need a requirement on a rate, are not covered.

3.2 Within-run floor

Production evidence comes as series of consecutive parts. Each series is cut into batches of B consecutive parts, and each batch is summarized by a robust center m (here the 10 % trimmed mean). The within-run floor of characteristic c, the floor for short, is defined as

$$F_c = Q_\alpha \{\, |m_i - m_j| : i < j \text{ batches of the same alternative} \,\}, \tag{3}$$

where Q_α is the α-quantile. Thus, the floor is the α-quantile of the difference between two batches of one design. Under a stationary normal model with a standard deviation σ_{b} between batch means and σ_{w} between parts, $F_c = 1.96\sqrt{2}\sqrt{\sigma_{\mathrm{b}}^2 + \sigma_{\mathrm{w}}^2/B}$ at $\alpha = 0.95$, which is the form of the repeatability limit (ISO, 2025, 1994). However, since the floor pools batch pairs up to 450 parts apart and grows with their lag, it is closer to a time-different intermediate-precision limit (ISO, 2023) of batch centers within a run. The reference protocol is $B = 50$, ten batches of a 500-part run, all pairs of batches and $\alpha = 0.95$. Every characteristic is scored in the floor of its own geometry, and the pooled floor is used only to convert distances to physical units. The floor rests on three assumptions: batches of one design differ only by drift and scatter within a run (A1);

future runs vary within themselves as the runs it is estimated from do (A2); and the measuring system is repeatable, with a resolution small against the floor (A3).

Four properties of the floor are important for its use. Firstly, it has the unit of the characteristic, so distances in floors put all characteristics and predictors on a common scale; however, equal distances do not mean equal nonconforming fractions. Secondly, it is a property of a protocol, and σ_{b} and σ_{w} are reported (`meas_floor`) so that it can be recomputed. Thirdly, it contains whatever drifts between batches, the process and the measuring system alike, so without a reference artifact scanned through the series it is a floor of production and measurement together, not of production alone. Finally, it excludes the variation between runs, which makes it a lower bound of the variation of series production. It should be noted that a new family, whose floor is unknown before it is produced, is scored in the floor of the produced family. Two alternatives a and b are resolvable for c when their effect-to-scatter ratio $\mathrm{ESR}_c(a,b) = |\bar{m}_a - \bar{m}_b|/F_c$ is at least one, where $\bar{m}$ is the center of all parts of an alternative. For a continuous parameter x with local sensitivity $s_c = \partial\bar{m}/\partial x$, the minimum resolvable change is $F_c/|s_c|$.

3.3 Decision rules

Eight decision rules are used for the main results, and they are listed in Table 2. The nominal simulation (M0) is the simulation at nominal sheet thickness and friction. The scaled simulation (M1s) fits $q_p = a + b\,s_0$ to the calibration alternatives by least squares, where s_0 is the nominal simulation; it is the scale term of a multi-fidelity correction without its discrepancy. The envelope rule (M2) adds the median offset of the calibration alternatives and their largest absolute residual to the top of the simulation envelope over the thicknesses and friction coefficients of the dataset. The GP-corrected envelope (M5) models the offset between q_p and the envelope as a GP (Rasmussen and Williams, 2006) fitted by maximum marginal likelihood. Its noise variance is bounded below by the variance between the calibration series of one geometry and force (the noise-variance bound). The nearest produced setting (NN) is the mean q_p of the calibration alternatives at the nearest force, which corresponds to a look-up in a capability database. Finally, the force trend (M1sn) is the least-squares line of q_p in force, the counterpart of M1s without the simulation, since within a geometry the simulation is a function of the force.

The trend GP (M7n) is a learner specified for six to nine produced alternatives. It is a GP of q_p with a linear mean in force and lubrication, whose coefficients are integrated out (universal kriging). Instead of hard bounds, weakly informative priors are used on the length scales and the noise. Its noise term is made of the known sampling variance of each q_p and an estimated between-run nugget, which absorbs no modeled effect. Its counterpart with the simulation (M7) is the autoregressive multi-fidelity model (Kennedy and O'Hagan, 2000), with the simulation in the mean and a GP discrepancy; with two calibration forces, its scale on the simulation acts as a force slope. In this study, the intervals are separated from the predictors: P+RM puts around the prediction of a rule P a relation-matched interval, the 90 % split-conformal quantile (Lei et al, 2018) of the absolute errors of P over the cases of the same relation in the other family, which is a group-conditional construction (Vovk et al, 2005).

Table 2 Decision rules of the main results (the further rules are defined in Section 3.3)

Rule	Prediction of q_p	Interval (level)	Evidence
M0 nominal simulation	simulation at nominal thickness and friction	point	none
M1s scaled simulation†	$a + b$ M0, least squares	point	q_p
M2 envelope + residual margin	envelope top plus the median offset	largest residual $(\frac{n-1}{n+1})$	q_p
M5 GP-corrected envelope	envelope plus a GP offset, ML-II	GP (nominal 0.9)	q_p
NN nearest produced setting†	mean q_p at the nearest produced force	point	q_p
M1sn force trend†	$a + b$ force, least squares	point	q_p
M7n trend GP‡	linear mean plus GP, MAP under priors	GP (nominal 0.9)	q_p
NN+RM‡	NN	relation-matched (nominal 0.9)	q_p

Fitted on the calibration alternatives; descriptors: geometry indicator, blank-holder force and lubrication rank. Pre-specified unless marked: † added after the first results; ‡ added in the revisions for this journal. Levels: guaranteed only with a sibling (largest residual, $n = 8$ calibration alternatives), whose held-out alternative is exchangeable with them; otherwise nominal.

The errors are borrowed because a family's own calibration set has no case of its new-setting relation (with a sibling, for one construction). The 21 (interpolated), 42 (extrapolated) or 63 cases behind a quantile come from one family and from one or a few shared fits. Therefore, they are not exchangeable with the cases of the other family, and the interval carries no coverage guarantee (Barber et al, 2023); its 90 % is nominal. Moreover, no interval from n exchangeable cases promises a miss rate below $1/(n + 1)$, so a guaranteed 5 % needs 19 produced alternatives in a relation.

In addition to M7, ten further rules complete the comparison. M1 adds the median offset of the calibration alternatives to M0, and M1n is their median q_p. M2n and M5n are M2 and M5 without the simulation. Mc fits one friction coefficient per lubrication pattern to the draw-ins by least squares, without a discrepancy, whereas Mcd calibrates the friction with a GP discrepancy (Kennedy and O'Hagan, 2001) and predicts by the posterior mixture over the friction grid. M3 corrects the matched simulation of every part by gradient boosting (Friedman, 2001), and M3n models the characteristic by boosting without the simulation; both have jackknife+ intervals (Barber et al, 2021). M0w is the range of the simulation over thickness and friction. Finally, M6, a robustness variant, puts a conformal margin on standardized residuals around the center of M5.

3.4 Decisive and safe distances

A rule is scored by judging each produced alternative as a new design at requirements placed at a known distance from the truth, as given in Eq. (4).

$$R = q_p(a, c) + d\, F_c, \tag{4}$$

where d is the requirement distance in floors. Thus, a is adequate exactly when $d \geq 0$. Let $\kappa(\delta)$ and $\omega(\delta)$ be the shares of correct and wrong verdicts among all feasible

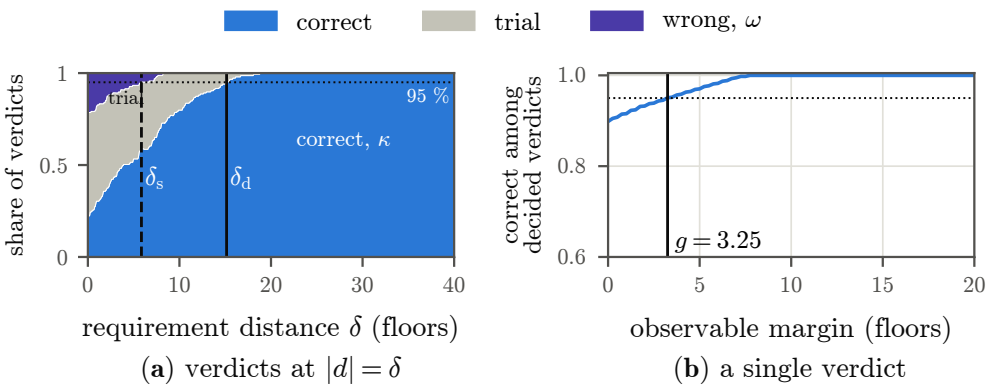


Fig. 1 (a) Verdict shares of M7n for an extrapolated setting against the requirement distance, with δ_d and δ_s, and (b) correct share of decided verdicts against the margin, with the in-sample pooled g

verdicts at $|d| = \delta$. Every case gives one verdict on an adequate alternative and, where the requirement is not negative, one on a failing alternative. The *decisive distance* δ_d is the smallest δ beyond which $\kappa \geq 0.95$, and the *safe distance* δ_s is the smallest δ beyond which $\omega \leq 0.05$, as shown in Fig. 1(a). In other words, for a rule that always decides, $\delta_d = \delta_s$ is close to the 90th percentile of its absolute error in floors. On the other hand, an interval rule trades decisiveness for safety: between δ_s and δ_d, it sends designs to trial rather than misleading. Both distances are points on operating characteristics over a population of design situations, and δ_d is the analogue of the indifference zone of a selection procedure (Bechhofer, 1954) at a probability of correct selection of 0.95.

The distances are exact up to their grid of 0.05 floor. However, they are order statistics, so a few verdicts far out can move them: with 126 cases, some 240 verdicts, the 95 % threshold rests on about a dozen. Distances below about one floor are reported as "<1", since production reproduces the decided quantile only to about one floor within a run (Section 5.1). The distances weigh false accepts and false rejects equally. Moreover, as δ grows, failing-side requirements become infeasible for characteristics close to zero, so the failing side carries half the weight up to a few floors and a third at about 100 floors. Therefore, the safe distances of each side are also reported.

3.5 Evidence relations and calibration budget

How a new design relates to the produced alternatives is part of the decision situation, and three evidence relations are distinguished. With a *sibling*, an alternative at the same process setting has been produced that differs only in the factor varied at a fixed setting. Here, this factor is the lubrication pattern, which the nominal simulation does not represent. A *new process setting* lies between produced settings (interpolation) or outside them (extrapolation). Finally, a *new family* shares no produced alternative. A team knows, before it decides, whether the produced siblings of a new variant differ resolvably from each other. Therefore, the sibling relation is stratified by this decision-time information, not by the production of the new design itself. The calibration budget is the dependence of the distances on the number k of produced alternatives.

3.6 Procedure for design-stage decisions

The framework is turned into the procedure of six steps that is shown in Fig. 2. Steps 1 to 5 are carried out once per design family and repeated when production evidence is added, whereas step 6 is carried out for every new design. Steps 1 to 4 frame the decision (family, alternatives, characteristics, form of the requirement and costs), measure the floor with a stated protocol, state which differences production resolves, and calibrate the candidate rules with and without the simulation.

Step 5 qualifies the rules of the family by their distances for each evidence relation, and it fixes a *guard band* g for each rule and relation. The guard band is the smallest observable margin between the predicted interval and the requirement beyond which at least 95 % of the decided verdicts are correct. Conditional on what is observed, g depends on the distribution of the requirement distances that the team expects, i.e., on its prior, as the global risks of conformity assessment do (Joint Committee for Guides in Metrology, 2012). The reference prior weights the 47 distances of the evaluation grid within ±20 floors equally (0, ±0.5 to ±10 in steps of 0.5, ±12, ±15 and ±20). It stands for requirements set a few to some tens of floors from the true 95th percentile; ±20 floors are about ±1 mm of mid-side draw-in, ±1.6° of wall angle and ±0.2 mm of cup depth. A release-like prior, the grid within ±5 floors, stands for requirements near capability limits. A team replaces both by the distribution of its own requirements, for example past drawing limits against achieved 95th percentiles. The band is estimated per family from the relations that its production exercises, and it is validated by leaving one alternative out. A relation that the production of the family cannot exercise, such as interpolation in a family produced at two forces, borrows the band of the same relation in another family, with the caveat of Section 3.3, or sends the design to trial. Stated costs may only widen the band, to the one minimizing Eq. (2) over the prior; therefore, they never lower the share correct below 95 %.

In step 6, the relation of the new design is established, and the rule qualified for it is applied (with stated costs, the cheapest one). The interval of the rule is widened by g floors, and the step returns meets or fails with the margin in floors, or a trial when no rule is qualified or the requirement lies inside the widened interval. In pseudocode, step 5 is: for each rule and relation, collect the held-out verdicts over the prior; for $g = 0, 0.25, \ldots, 20$, compute the share correct among verdicts decided with a margin of at least g; take the smallest g beyond which it stays at or above 0.95 and, with stated costs, raise it to the g that minimizes Eq. (2). Step 6 is: $V =$ meets if $U + gF \leq R$, fails if $L - gF > R$, and trial otherwise.

3.7 Worked example

A process planner has to decide whether the concave cup of Section 4.1 can be run at 300 kN with medium lubrication without a trial run. This setting has not been produced yet, whereas the six alternatives at 100 and 500 kN have been produced. In step 1, the decision is framed: the characteristic is the mid-side draw-in, by which the blank-holder force is set, and the requirement, chosen for illustration, is that 95 % of the parts draw in by at most 15.30 mm. In addition, a trial costs a twentieth and a false reject half of a false accept. In step 2, the floor is obtained from the six produced

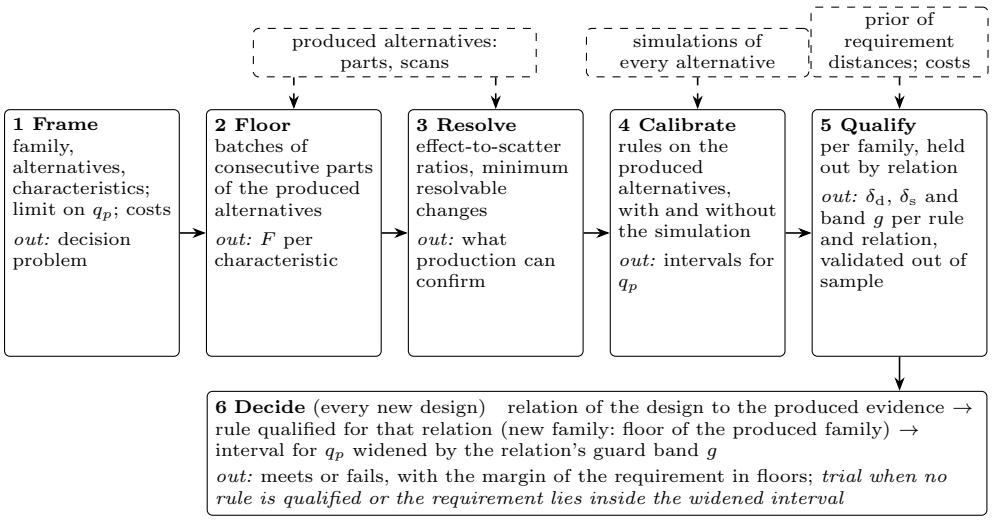


Fig. 2 Steps of the procedure: steps 1–5 once per design family and step 6 for every new design (dashed boxes: the evidence and inputs that the steps use)

series as 0.041 mm; the draw-in is quantized in steps of 0.019 mm, and three values hold a median 89 % of a series. Step 3 shows that a force change of about 40 kN is resolvable between 100 and 500 kN. In step 4, the rules are calibrated on the six alternatives. In step 5, no rule can be qualified for interpolation on this family, since its production holds no interpolated case. Therefore, the band is borrowed from the convex family, which was produced at all three forces, on the assumption that the two families err alike. There, the nearest produced setting decides at 5.3 floors and needs a band of 2.5 floors, which the stated trial cost widens to 5.0 floors. In step 6, the nearest setting, the mean of the six alternatives at the neighboring forces, predicts 14.99 mm, a margin of 7.4 floors, which is more than the band; therefore, the planner releases the setting. The trend GP agrees with a margin of 5.4 floors. On the other hand, the intervals of M2, M5 and M5n contain the limit and would send the design to trial, and the nominal simulation, qualified for no relation, predicts 15.66 mm. The 500 parts had a 95th percentile of 14.88 mm, 10.0 floors below the limit. How often such verdicts are right is reported in Section 5.5.

4 Application to a deep-drawing process

In this section, the produced alternatives, the simulations and the quality characteristics of the deep-drawing process are introduced first. After that, the evaluation design and the computation are explained.

4.1 Produced alternatives and production data

The production data are the Real Deep Drawing and Cutting dataset (RDDAC) of the University of Stuttgart (Baum and Heinzelmann, 2026; Baum et al, 2026a). In this

dataset, square blanks of DP600 dual-phase steel, about 1 mm thick, were deep drawn into cups (OP10) and cut (OP20) on an industrial press with one tool set. The blanks are taken as 210 mm wide, the blank of the simulation dataset (1 mm of width moves the measured draw-in by 0.5 mm, i.e., 10 floors). The two die geometries of the tool set, with concave and convex side walls, differ in base curvature, wall angle and bottom radius at once. Each geometry was run at three blank-holder forces (100, 300 and 500 kN) and three lubrication patterns applied to the blank (coarse, medium and fine, with series medians of 1.15–1.48, 0.92–1.23 and 0.85–1.13 $\mathrm{g/m^2}$ of oil). Thus, there are 18 alternatives, each produced as one series of 500 consecutive parts. For every part, four load-cell signals at 300 Hz, the punch position and temperature, sheet-thickness and oil-film traverses, and laser scans after both operations are available. The series warm up by 1.0–5.0 K over their first 150 parts from punch temperatures of 20.5–29.6 °C, and their median sheet thickness is 0.986–0.993 mm. Moreover, the 100 kN series ran at 47 mm/s against 78–79 mm/s; therefore, the extrapolation to 100 kN also changes the sliding speed and the stroke rate on which the lubricant film depends. The experiments were collected between September and November 2025, in contiguous blocks of 500 parts per alternative. However, the documentation states neither the order of the blocks nor whether the parts were scanned in production order, and it gives no blank size, material certificate, coil record, tool drawing, press or cushion type, or how the lubrication was applied.

4.2 Simulation dataset

The simulations come from the Deep Drawing and Cutting Simulations dataset (DDACS) (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025a). It is a generic dataset of some 32,000 LS-DYNA simulations of the same cup family, and it was neither set up for nor validated against the produced process. According to its documentation, the tools are rigid, and the blank is modeled by shell elements with seven integration points through the thickness and nodes about 1 mm apart, with a scaled DP600 flow curve and constant Coulomb friction. The drawing depth is 30 mm, the trimming in OP20 is modeled without cutting tools, and springback is computed after both operations. On the other hand, the documentation leaves open the element formulation, whether the contact accounts for the shell thickness, the yield criterion and hardening law, the blank-holder actuation and the springback procedure, and the stroke speed is not represented. For the RDDAC tools, 396 simulations vary the sheet thickness (0.95–1.00 mm) and the friction coefficient (0.05–0.15) at each geometry and force, all with the nominal flow curve. A part is matched to the simulation with the nearest thickness and with the friction coefficient that the linear mapping of the dataset assigns to its oil film, an assumption of the authors of the dataset on which M3, M3n and the matched simulations depend. The nominal simulation is at 0.99 mm, nearest the median thickness, and at friction 0.10, the middle of the grid. It should be noted that the simulated characteristics carry extraction noise: the two directions of the quarter model differ by 1.8–2.3 floors in wall angle, and the dome is rough along the friction grid by 2.4–9.8 floors (standard deviations; `meas_sim`).

4.3 Measurement and quality characteristics

The parts were measured with a single-sided laser line scanner, with the cup bottom facing the scanner (Fig. 3). The scanner records a height and a luminescence image of 3200 by 2000 points, which are 0.077 mm apart along the laser line (x) and 0.158 mm apart between lines in the direction of motion (y), with height steps of 7.75 µm. However, the scales are not traced to the part. The lateral scale comes from the scanner specification, and it was checked against a 9 mm wide calibration block that also gives the height (1 mm), against a nominal repeatability of 5 µm. The line spacing, which the corner draw-in, the arm angle and the dome use but the mid-side draw-in, the wall angle and the depth do not, was estimated from the scans and corrected by a fit of the scale of the simulation. Therefore, the absolute values are uncalibrated. This limits every absolute comparison with the simulation, since a lateral scale error of 1 % moves the mid-side draw-in by 0.9 mm, i.e., by 18 floors. However, it does not limit the floor, which is a difference within an alternative that a constant scale error only rescales.

Seven characteristics were extracted with the same definitions from the scans and from the simulated nodes, referred to the plane of the cup bottom, and they are listed in Table 3. The mid-side and corner draw-in and the flange waviness are process indicators on the flange, which the cut removes. The wall angle and the cup depth are measured after drawing, because the cut removes the flange that defines the depth and the cut walls facing the scanner read several degrees too steep. Finally, the arm angle and the bottom dome are measured on the cut part, and their requirements are on absolute values. The four sides of a part measure several characteristics twice, which exposes direction-dependent systematic errors of up to 3.5 mm in flange width and 10° in wall angle (Table 3). The directions kept in this study avoid the worst of them, but their residual bias is unknown. In addition, the mid-side draw-in is quantized in steps of 0.019 mm (0.38 floors), so a series takes only 6 to 21 values, as seen in Fig. 4.

4.4 Evaluation design

Every produced alternative is judged as a new design under grouped cross-validation (Roberts et al, 2017), with one scheme per evidence relation. For a *new variant*, the alternative alone is held out, and the other eight of its geometry calibrate the rules, two siblings at its force among them. For a *new process setting*, the three alternatives of one force of a geometry are held out, and the six others calibrate; 300 kN is interpolated, whereas 100 and 500 kN are extrapolated. For a *new family*, one geometry calibrates the rules for the other (two transfers), and a few-shot variant adds one to three produced alternatives of the new family. Every calibrated rule is evaluated with and without the simulation (M1s/M1sn, M2/M2n, M3/M3n, M5/M5n and M7/M7n). Each relation has 126 cases (18 alternatives and 7 characteristics). However, the units of replication are the 18 alternatives, whose characteristics are correlated.

The intervals of Table 5 are obtained by resampling the held-out alternatives within each geometry 1000 times (those held out together for a new setting individually), with the fitted rules held fixed. In each resample, every true q_p is replaced by a block-bootstrap replicate (non-overlapping blocks of 50 parts). Since the resamples are common to all rules, they give paired intervals of differences, which are descriptive and

Table 3 Quality characteristics and their measurement by the laser scanner

Characteristic	Op.	Region; definition	Step/F	Values	Direction-dependent differences
Draw-in, mid-side	OP10	flange mid-sides; $(210 - W_x)/2$, W_x along the laser line	0.38	6–21	$W_y - W_x$ 2.29–3.47 mm; W_y, in the direction of motion, scatters 3.7–17.7 times more
Draw-in, corner	OP10	flange corners; mean of the two diagonals	<0.01	491–500	the two diagonals differ by 1.64–2.17 mm
Flange waviness	OP10	flange; residual SD about a quadratic surface	<0.01	469–494	–
Wall angle	OP10	side walls at mid-depth; mean of east and west	<0.01	497–500	convex: north wall −1.64 to −0.92° from east/west (concave −0.05 to +0.39°); south wall shadowed, 6.7–10.0° off; after the cut, north 3.3–5.6° off (convex)
Arm angle, cut	OP20	cut arms, center lines; mean of the four arms	<0.01	496–500	–
Cup depth	OP10	flange above the bottom plane; median	<0.01	486–497	the flange plane is cut away at OP20
Bottom dome, cut	OP20	bottom sag at $r = 40$ mm; paraboloid fit	<0.01	474–497	–

Op.: operation measured (OP10 after drawing, OP20 after cutting). Step/F: median step between adjacent values in one series over the floor; Values: distinct values per series. Differences from the redundancy of the four sides, ranges of the series medians; the directions used avoid them, their residual bias is unknown. Absolute values are not calibrated against a reference at the size of the part.

not adjusted for the many comparisons. For the coverage, cluster-bootstrap intervals over the alternatives are computed where each alternative has its own calibration set. Each truth is the 95th percentile of a single run. Therefore, a sensitivity analysis adds to every truth a between-run component of standard deviation τ floors, from near-replicate reproducibility up to the variation between series. The selection of rules and bands is cross-fitted between the geometries: they are chosen on one geometry and evaluated on the other. Like any cross-validated estimate, the distances describe the fitting procedure on this design rather than one fitted rule (Bates et al, 2024).

4.5 Computation and development record

The constants of the floor and of the requirement grid were set before the decision results were seen; however, the archive cannot verify this, since there is no timestamped snapshot. The rules were not fixed in advance. M0–M5 and M3n were defined first, and everything else was added after the first results, most of it in response to simulated reviews of earlier versions, which a large language model produced in seven rounds. In four rounds, M0w, Mc, M1n, NN, M2n, M5n, the grouped schemes, the exact estimator, the noise-variance bound, M1s, M1sn, the jackknife+ intervals, the guard bands with their priors and the scoring of the new family in the floor of the produced family were added. In a fifth round, on the version prepared for this journal, M7n, M7,

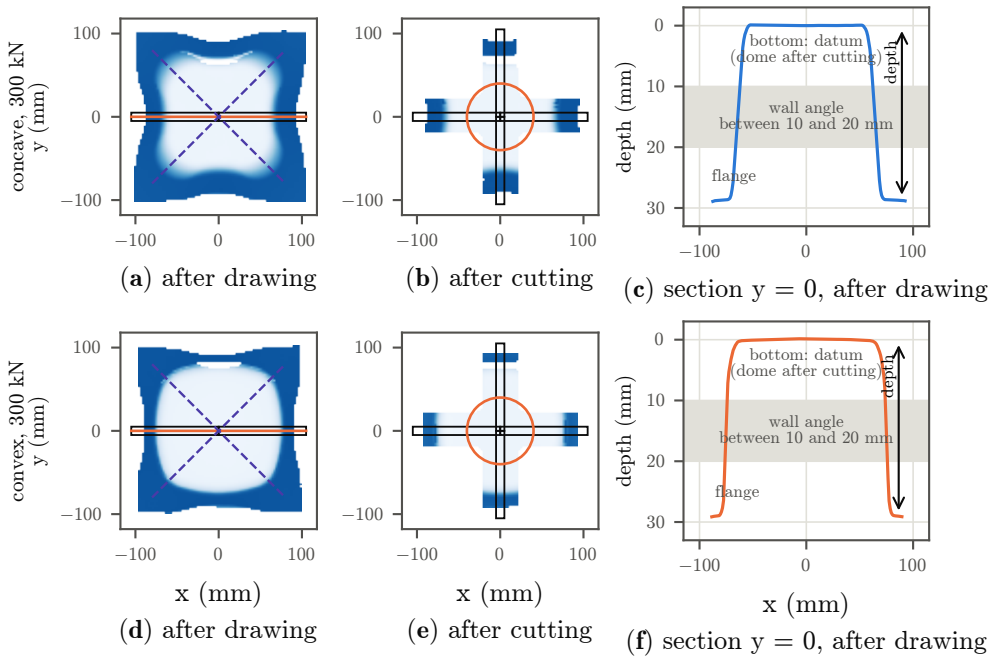


Fig. 3 Scanned parts after drawing and cutting: (a, b, d, e) depth below the cup bottom and (c, f) sections (lines and boxes: directions and regions of the characteristics)

Mcd, the relation-matched intervals, the decision-time strata, the costs, the between-run sensitivity, the cross-fitting, the few-shot transfer and the design-level verdicts were added. In a sixth and a seventh round, on its revisions, the guard bands available at decision time, the band widened by stated costs, the prior sensitivity of M7n, the trial under between-run variation and the costs under the rule of step 5 were added. Therefore, the comparison of rules is exploratory, and it carries the optimism of a selection made on the evaluation data (Cawley and Talbot, 2010; Varma and Simon, 2006), which the cross-fitting estimates. Every variant tried is reported in the paper or in the archived result files (Canseven, 2026), which are named in typewriter type in the text.

The computational details are as follows. M5 and M5n are GP regressions with an anisotropic squared-exponential kernel and white noise, fitted by maximum marginal likelihood on a geometry indicator, the force in 100 kN and the lubrication rank. M7n, M7 and Mcd work in pooled floors under log-normal priors with medians of 200 kN and one pattern step for the length scales, and two and one floors for the GP and nugget standard deviations; Mcd has a uniform prior over the 11 friction values of the grid. M3 and M3n use histogram gradient boosting with 500 incoming conditions drawn from calibration parts of the same pattern. The software used is Python 3.11, SciPy and scikit-learn (Pedregosa et al, 2011), and every setting is given in the archive.

5 Results

In this section, the floor of the process and the differences that production resolves are given first, and the simulation dataset is compared with production. After that, the distances of the rules, the learned rules and the guard bands are presented. Finally, the costs, the release-level requirements and the robustness of the results are examined.

5.1 Within-run floor of the process

The within-run floors of the seven characteristics are given in Table 4. According to the table, two batches of 50 consecutive parts of one alternative differ by up to 0.051 mm in mid-side draw-in and 0.078° in wall angle in 95 % of the batch pairs. Thus, one floor is 0.7 to 1.7 long-term standard deviations of the parts. Fig. 4 shows the 500 parts of one run in production order. The floors are set by drift within the run: without the time order, they shrink by a factor of 2.1 to 4.5, batches one position apart differ by 0.51–0.80 floors, and batches nine apart by 1.1–1.5 floors. Without the warm-up of the first 150 parts, the floors fall to 0.75–1.00 of the reference. The reference keeps the warm-up, because series production starts up after every break. Independent part-to-part noise, including the repeatability of the scanner, accounts for 15–33 % of each floor. Therefore, A3 holds for repeatability and, except for the mid-side draw-in, for resolution, while bias, linearity and stability remain untested.

Whether the drift comes from the process or from the scanner cannot be settled without a reference artifact. The candidates are the warm-up of the tools, lubricant building up on die and blank holder, and material and blank size varying along the stack. However, no recorded process signal explains the drift: the batch centers of force, punch temperature, thickness and oil film predict it out of fold with R^2 of -0.23 to 0.20. Moreover, the batch centers of the flange width and of the cup width at 15 mm depth, which punch and die largely fix, are only weakly correlated within the runs ($r = -0.27$ and $+0.27$). This bounds a common lateral scale drift of the scanner below 0.05 %, about 0.9 floor of mid-side draw-in, whereas drift in height and of the edge detection remain untestable. However, the variation between runs is not in the floor. The three lubrication series of one geometry and force differ by 1.2 floors (waviness) to 9.8 floors (dome) at the 95th percentile of their center differences, including the lubrication effects. In contrast, within a run, the 95th percentiles of two batches of 100 parts differ by 0.92–1.49 floors in 95 % of the pairs.

5.2 Resolution of production and comparison with the simulation dataset

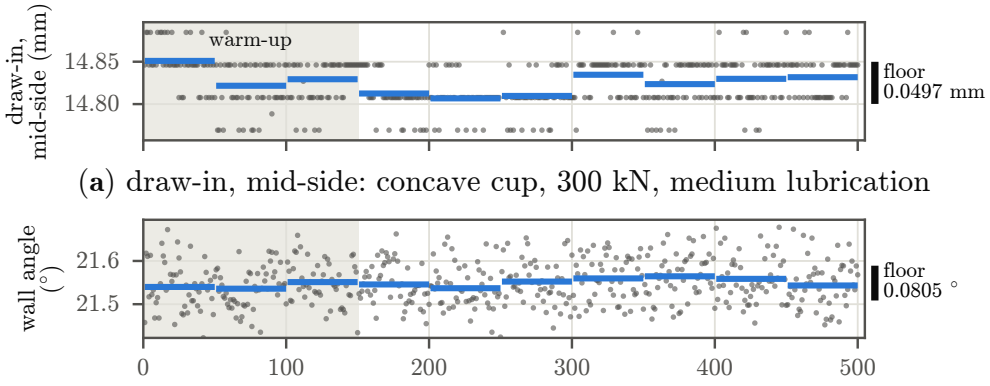
All 63 geometry contrasts exceed the floor, and 105 of the 126 force contrasts do. However, only 51 of the 126 lubrication contrasts exceed it, so a lubrication variant is often a near-replicate of its siblings. The minimum resolvable force change is 21–67 kN between 100 and 300 kN for most characteristics, and it exceeds either force interval for the wall angle and the dome.

The simulation dataset does not contain the produced process. This is shown at every friction and thickness of the grid by the three comparisons that depend neither

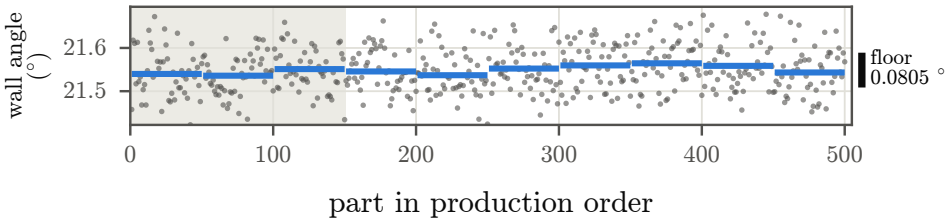
Table 4 Within-run floors of the quality characteristics, pooled and per geometry, with their spread over the runs

Characteristic	F	Interval	Concave/convex	Single runs	No warm-up	F/σ_{LT}
Mid-side (mm)	0.051	0.042–0.061	0.050/0.052	0.36–1.62	0.99	1.0/1.4
Corner (mm)	0.046	0.032–0.053	0.053/0.030	0.27–1.62	0.95	1.4/1.2
Wavin. (mm)	0.0028	0.0021–0.0050	0.0037/0.0021	0.21–2.08	0.89	1.6/0.7
Wall (°)	0.078	0.054–0.10	0.081/0.078	0.30–1.74	0.80	1.3/0.7
Arm (°)	0.11	0.089–0.15	0.12/0.100	0.40–2.16	1.00	1.7/0.8
Depth (mm)	0.011	0.0090–0.012	0.0087/0.012	0.39–1.45	0.75	1.4/1.6
Dome (mm)	0.0070	0.0052–0.0084	0.0034/0.0082	0.42–1.61	0.93	1.2/1.6

Batches of 50 consecutive parts, 95 % quantile of the difference of their 10 % trimmed means. Interval: 95 % over 2000 resamples of whole alternatives (the narrower interval over resampled batches is in the archive). Single runs: floor of one run over that of its geometry, range over the 18 runs. No warm-up: floor without the first 150 parts over the reference. σ_{LT}: standard deviation of the parts of a series (concave/convex); σ_{b}, σ_{w} and the drift ratio in `meas_floor`.



(**a**) draw-in, mid-side: concave cup, 300 kN, medium lubrication



(**b**) wall angle: concave cup, 300 kN, medium lubrication

Fig. 4 The 500 parts of one run in production order with the centers of its ten batches of 50 parts (bars) and the floor for (a) mid-side draw-in and (b) wall angle (warm-up shaded)

on the assumed blank nor on the scales of the scanner. Firstly, the simulated force effect on the mid-side draw-in is 2.8 to 7.4 times the measured one. Secondly, the cut arms of the concave cups bend downward in 89–100 % of the simulations but upward in every part, whereas the convex arm angles agree within 0.6°. Thirdly, the dome ranks the geometries in reverse in 76 % of the simulations. The levels, which depend on the blank and the scales, disagree as well if both hold (`sim_validation`). At the nominal thickness, the measured center lies within the range spanned by the friction grid in only 11 of the 42 combinations of characteristic, geometry and force. The simulated mid-side draw-in exceeds the measured one at 100 kN for every friction coefficient (by 1.8–2.0 mm at friction 0.10), and it comes within 0.011 mm of it at 300 kN only at the edge of the grid on the concave cup. Similarly, the corner draw-in exceeds the measured one everywhere, by 2.6–3.5 mm at friction 0.10 and by 1.3–3.3 mm at 0.15. Calibrating the friction cannot repair this within the grid. All 72 within-family

least-squares calibrations of Mc end at the upper edge of the grid (0.15), and with a discrepancy term, the posterior of Mcd piles up at the lower edge (0.05), where the simulated force effect is weakest. Furthermore, the mapping of the dataset from oil film to friction implies 0.8–3.1 mm of mid-side draw-in per g/m^2 of oil. This is an order of magnitude more than the slopes within the runs and between the lubrication patterns (at most 0.09 and 0.39 mm per g/m^2), even allowing for noise in the oil readings. Through this mapping, an envelope over the measured incoming variation varies with the lubrication pattern more than production does, which makes M5 less decisive with a sibling (24 against 7.2 floors). In addition, a GP surrogate of the simulations decides as the simulation does (109 against 108 floors).

The cup depth is simulated 0.70–1.06 mm (2.5–3.7 %) deeper than measured, and the wall at 15 mm depth, which is set by punch and die, is 2.0–2.2 mm (2.8–3.5 %) smaller in radius. The depth offset is neither definitional, since the bottom and the flange are seen from the same side of the sheet in both, nor due to the flange inclination, which would make the measured depth larger. A drawing depth 1 mm shorter would move the wall by only 0.2–0.4 mm, and a scale error would give an offset proportional to the radius (64 and 75 mm), not 2.0–2.2 mm at both. Another tool geometry fits best, but it is unconfirmed. Without the offset, M0 decides at 65 instead of 108 floors.

5.3 Decisive and safe distances by evidence relation

The decisive and safe distances of the rules are given in Table 5, and the verdict shares of NN, M5 and M7n are shown in Fig. 5. As seen in the table, the nominal simulation decides at 108 floors, and it errs mostly by false rejects.

For a new variant with a produced sibling, every calibrated rule improves on the nominal simulation, and the production record is the most decisive. The nearest setting, the mean q_p of the two siblings, is right beyond 3.4 floors (2.0–4.5), and the force trend beyond 3.5 floors. On the other hand, M5 and M7n decide at 6.8–7.2 floors, but they are safe below one floor. When stratified by what a team knows at decision time, the nearest setting needs 1.60 floors when the two produced siblings do not differ resolvably and 4.40 floors when they do. In physical units, the nearest setting is right beyond 0.10 mm of mid-side draw-in, 0.26° of wall angle and 0.04 mm of cup depth (Table 6). It is important to note that one produced sibling suffices: with a single alternative at the force of the new design, the nearest setting decides at 3.5 floors (2.3–4.6), and further alternatives do not improve it, while they make M5 more decisive and safer.

For a new process setting, with all alternatives at the new force held out, the force trend and the nearest setting decide at 8.2 and 8.5 floors, respectively, whereas the interval rules decide only at 15–19 floors, although they are safe at 5–8 floors. Interpolating is much easier than extrapolating (the nearest setting at 4.7 against 9.5 floors). However, at 300 kN, the midpoint of a balanced calibration set, the nearest setting, the force trend and the center of M5n coincide by construction. Therefore, interpolation cannot separate the mean-type rules. When extrapolating a fitted trend, the scaled simulation falsely accepts failing alternatives up to 165 floors from their limit, on the cup depth. On the other hand, it is more decisive than the force trend on

Table 5 Decisive and safe distances of the rules (floors) for each relation of the new design to the produced evidence

	Sibling		Interpolated		Extrapolated		New family	
Rule	δ_d/δ_s	FA/FR	δ_d/δ_s	FA/FR	δ_d/δ_s	FA/FR	δ_d/δ_s	FA/FR
M0	108	17/109	109	17/115	106	12/109	104	28/105
M1s†	4.9	4.7/4.9	19	30/19	19	165/7.9	121	125/118
M2	23/<1	<1/<1	18/2.7	3.7/2.4	18/9.9	8.7/10.0	41/26	27/9.7
M5	7.2/<1	<1/<1	19/2.8	2.4/3.1	19/8.9	8.4/9.7	35/31	33/15
interval of δ_d	6.6–8.4		18–20		18–20		31–39	
M7n‡	6.8/<1	<1/<1	7.4/3.4	2.2/3.4	15/5.9	7.5/<1	132/130	129/167
interval of δ_d	6.0–7.3		7.1–8.1		15–16		131–132	
NN†	3.4	2.8/3.4	4.7	3.1/4.7	9.5	11/5.6	130	129/168
interval of δ_d	2.0–4.5		4.3–5.1		6.6–11		129–131	
M1sn†	3.5	3.5/3.6	4.7	3.1/4.7	8.9	9.3/7.4	131	129/168
NN+RM‡	6.3/<1	<1/<1	9.6/<1	<1/<1	20/<1	1.5/<1	293/4.5	4.1/44
Mcd‡	7.2/<1	<1/<1	22/1.6	<1/2.5	24/5.8	6.6/2.6	30/19	29/12
Trial, first 20 parts	1.4	1.5/1.2	1.4	1.5/1.2	1.4	1.5/1.2	1.4	1.5/1.2
Trial, first 50 parts	0.90	1.0/0.85	0.90	1.0/0.85	0.90	1.0/0.85	0.90	1.0/0.85

Over all seven characteristics; one number for rules that always decide; <1: below one floor. FA/FR: safe distances on the false-accept and the false-reject side. 126 cases with a sibling and for a new family, 42 interpolated, 84 extrapolated; a new family in the produced family's floor (in its own floor, and all rules, in `dec_resolution` and `learn_resolution`). Interval: 95 % over 1000 resamples of the held-out alternatives, fits fixed. Mcd: friction calibration with a discrepancy term (Section 3.3). Trial: the first n parts of the design's own run, a within-run lower bound. †, ‡: added after the first results and in the revisions for this journal.

the mid-side draw-in and the arm angle when extrapolating. The multi-fidelity model M7 fails in the same way, with false accepts up to 118 floors on the cup depth.

For a new family, scored in the floor of the produced family as the procedure prescribes, every rule without the simulation fails (130–135 floors), since nothing in the record of one family describes the other. With the simulation, Mcd decides at 30 floors and M5 at 35 floors, while M3 is safe at 17 floors and M2 at 26 floors. Per characteristic, M5 decides at 5–10 floors for the draw-ins and the wall angle, which the forming model represents directly, but at 20–40 floors for the depth, waviness, arm and dome. Nevertheless, production knowledge still carries over where a drawing nominal transfers: for the wall angle, the design angle plus the deviation of the produced family from its own design angle decides at 1.9 floors, against 7.5 floors for M5. However, two transfers do not resolve which rule is the safest.

5.4 Learned rules and their intervals

The GPs fitted by maximum marginal likelihood reduce to the baselines in this design (`learn_gp_diagnostics`). For a new setting, the force length scale and the noise variance of M5n sit at their bounds in most fits, and its center lies within 0.01 floor of the calibration mean in 93–95 % of the cases; with a sibling, it lies a median 0.28 floor from the nearest setting. The trend GP, specified for this regime, reduces to the force trend instead. Its center lies a median 0.33–0.36 floor from the force trend for a new setting, and its force length scale, which two force levels cannot inform, stays near the

Table 6 Decisive distances per characteristic in floors and, in brackets, in the unit of the characteristic

	NN			M1sn	M1s	M5	
Characteristic	Sibling	Interp.	Extrap.	Extrap.	Extrap.	New family	P_{pk}
Mid-side (mm)	2.0 (0.099)	3.8 (0.19)	6.5 (0.32)	5.7 (0.28)	4.0 (0.20)	5.5 (0.27)	1.3
Corner (mm)	4.5 (0.13)	1.9 (0.058)	6.7 (0.23)	2.8 (0.083)	2.9 (0.100)	9.9 (0.31)	2.5
Wavin. (mm)	<1 (<0.0028)	4.8 (0.014)	11 (0.040)	9.3 (0.027)	21 (0.077)	22 (0.047)	0.9
Wall (°)	3.4 (0.26)	<1 (<0.078)	1.9 (0.15)	1.8 (0.14)	2.0 (0.16)	7.5 (0.58)	1.7
Arm (°)	2.8 (0.28)	5.9 (0.59)	5.4 (0.53)	9.5 (0.94)	5.9 (0.69)	39 (3.9)	1.6
Depth (mm)	3.3 (0.039)	4.4 (0.038)	15 (0.13)	9.9 (0.086)	165 (1.4)	20 (0.20)	2.4
Dome (mm)	7.0 (0.038)	5.3 (0.044)	4.3 (0.020)	3.0 (0.020)	15 (0.056)	40 (0.21)	4.0

18 cases per characteristic with a sibling and for a new family, 6 interpolated, 12 extrapolated; a new family in the produced family's floor. In brackets: the decisive distance computed in the unit, every case scored in the pooled floor (not the distance in floors times a floor). P_{pk}: median performance index of a requirement at the decisive distance of NN with a sibling. Both distances of every main rule in `qual_per_characteristic`.

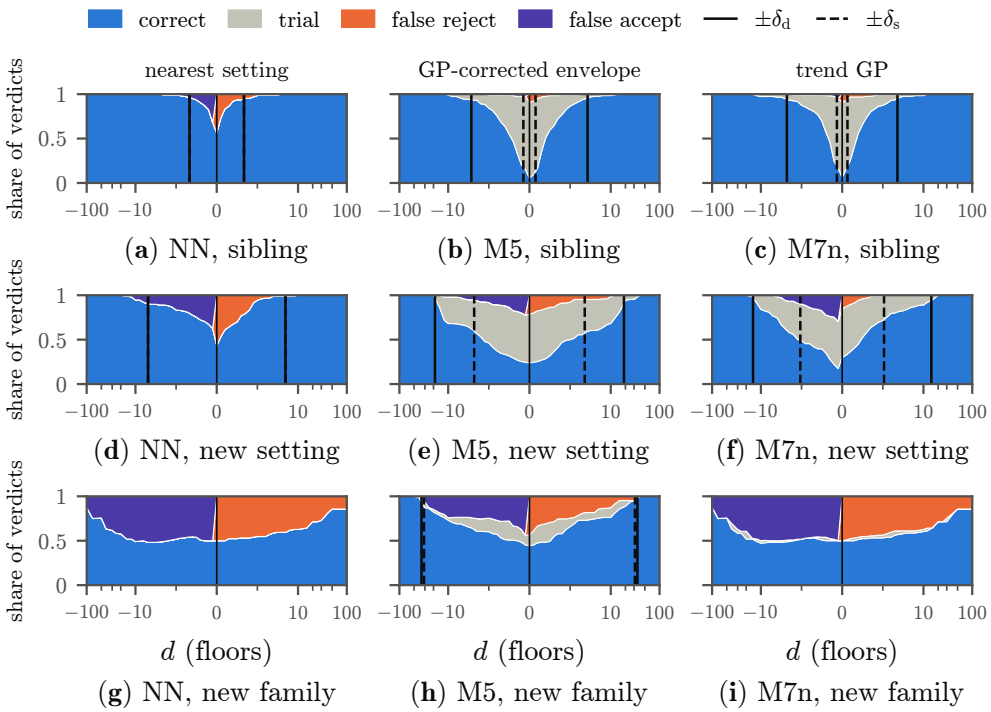


Fig. 5 Verdict shares of NN, M5 and M7n against the requirement distance d by relation (symmetric logarithmic axis, linear within ±10 floors; new family in the produced family's floor; 126 cases each)

prior median of 200 kN. Moreover, its between-run nugget, 0.4–0.5 floors, is three to four times the sampling error of the truths. The trend GP is more decisive than M5n for a new setting (14.5 against 17.2 floors) and as safe, but it is less decisive than the nearest setting in every relation (by 3.5 floors with a sibling and 6 floors for a new

setting). Its intervals cover only 52 % when interpolating and 56 % when extrapolating (17 and 33 % on the mid-side draw-in, and none on the waviness). This is because their width is learned from the variation between lubrication patterns at produced forces, which is sibling-type information. Halving or doubling any of its four prior medians moves its decisive distances by at most 1.4 floors, keeps it within 0.4 floor of the force trend and keeps its coverage for a new setting at 48–62 % (`learn_m7n_priors`). With the simulation in its mean (M7), it gets worse for a new setting (28 floors), because the simulated force trend enters it. Tuned part-level boosting decides no closer than 11.2 floors with a sibling and 13.1 floors for a new setting.

The contributions of the prediction and of the interval of a rule can be separated. A relation-matched interval borrowed from the other family makes even the nearest setting safe for a new setting (safe distance <1 floor, against 6.8 floors for M5 and 5.2 floors for M7n), at the decisive distance of the learned intervals (19.5 against 18.7 and 14.5 floors). The safety comes from the borrowed evidence, not from learning (`learn_resolution`). Around the centers of the trend GP and the force trend, the same interval decides closer when extrapolating (17.6 and 17.8 against 20.1 floors), whereas around simulation-based centers, it does not make the rule safe (false accepts up to 151–155 floors). Furthermore, a trend GP calibrated on both families, which borrows through a family term, is safe at 2.75 floors. The coverage of NN+RM, 88 % when interpolating and 93 % when extrapolating, comes from two family swaps, not from a guarantee: it ranges from 76 to 100 % between the families and from 67 to 100 % between the characteristics (`qual_coverage`). Around production-record and trend centers, the relation-matched intervals have the best interval scores (Gneiting and Raftery, 2007) for a new setting (`qual_coverage`). With a sibling, the GP intervals cover 83–86 % (cluster intervals 0.72–0.94), which is compatible with their nominal 90 %, and the jackknife+ intervals of M3 cover 96 % (0.93–0.98) against a guarantee of 0.78. For a new family, no interval covers well except at widths that send every design to trial. With one to three produced alternatives of the new family, its own nearest setting or its offset to the record of the other family decides at 5.9–10.5 floors in the floor of its nine runs, which a team with so few alternatives does not have.

5.5 Guard bands at decision time

In step 5, the margins are turned into guard bands as a team could estimate them at decision time, and the results are given in Table 7. With a sibling, the band comes from the sibling cases of the family itself. For a new setting, it comes from the same relation in the other family, since a family produced at two forces has no interpolation case of its own. Under the reference prior, the nearest setting keeps 95 % of its decided verdicts correct in each relation, pooled over the two families (92–100 % per family). M2 and M5 send 52–57 % of the verdicts to trial when interpolating, and when extrapolating, they qualify only on the convex cup, whose band gives 83–84 % correct on the concave cup. However, the 95 % is an average over the characteristics: the nearest setting is right in 89 % of its decided verdicts on the dome with a sibling, in 88 % on the waviness when interpolating and in 85 % on the cup depth when extrapolating (`qual_guard_decision`). In addition, the bands are within-run: with a

Table 7 Guard bands available at decision time (floors) with their share correct and trial rate

	Sibling			Interpolated			Extrapolated		
Rule	g	Correct	T	g	Correct	T	g	Correct	T
Reference prior									
NN†	0/0.5	95	2	2.5/1.5	95	18	5.75/9.5	95	58
M1sn†	0.5/1.0	95	7	2.5/1.5	95	18	6.5/5.5	95	47
M2	0	99	54	0.5/0.75	95	52	6.25/none	84	81
M5	0	99	23	0/1.75	95	57	5.0/none	83	80
M7n‡	0	98	19	1.75/0	95	28	4.75/2.75	96	60
NN+RM‡	0	99	28	0	99	42	0/0.5	97	73
Release-like prior (±5 floors)									
NN†	1.0/3.0	96	41	4.75/3.5	93	67	8.5/14.5	75	92
M1sn†	2.25/3.0	95	47	4.75/3.5	93	67	8.5/8.25	94	84
M2	0	97	76	3.25/2.25	94	85	10.25/none	72	91
M5	0/6.0	95	67	1.5/3.25	88	84	9.25/none	67	91
M7n‡	5.5/7.25	84	91	4.25/1.75	87	74	7.0/6.25	93	90
NN+RM‡	0/1.0	96	72	0.5/0	95	77	0/5.25	75	93

g: band of the family's own sibling cases with a sibling; for a new setting, which a family produced at two forces cannot exercise, the band of the same relation in the other family (concave/convex; one number when equal; none: no band up to 20 floors reaches 95 %). Correct: share correct among decided verdicts (%), pooled over the two families; with a sibling leaving one alternative out within its family (refitting the band's cases without it, NN keeps 96 and 95 % correct and sends 3 and 52 % to trial), for a new setting the other family's band applied to this family. T: share of verdicts sent to trial (%). No rule qualifies for a new family. Shares per family, verdict counts and the characteristics in `qual_guard_decision`.

between-run component of $\tau = 3$ floors, the pooled bands of the nearest setting grow from 0.25, 2.0 and 7.0 to 2.9, 4.0 and 8.9 floors (`qual_between_run`).

The bands depend on the prior as much as on the rules (`qual_guard_prior_width`). Under the release-like prior, the bands widen, and the nearest setting keeps only 75 % of its decided verdicts correct when extrapolating. Under the same prior, with a band estimated from the cases of both families except the held-out force cell, the nearest setting and NN+RM reach 0.72 and 0.73 when extrapolating. On the other hand, when interpolating, where only the cell of the other family remains, M5 and M7n keep the 0.88 and 0.87 of Table 7 (`qual_guard_oos`). Under the reference prior, no rule qualifies for a new family, which agrees with the practice of trying out every new die: Mcd reaches 95 % on one transfer with a band of 18 floors, but that band gives 73 % on the other.

5.6 Costs, trials and release-level requirements

Fig. 6 shows the expected cost of the verdicts by Eq. (2), with a false reject costing half a false accept and with the band of each rule chosen as step 5 prescribes, from its 95 % band upward. The following results hold under both priors and for every ratio of false-reject to false-accept cost (`qual_cost_cheapest`). For a new setting, the nearest setting or the force trend is the cheapest rule at every trial cost. With a sibling, the nearest setting is the cheapest once a trial costs a tenth of a false accept or more. Below that, the cheapest rules differ by thousandths of a false accept, which this design does not resolve. For a new family, no rule qualifies, so every design goes to

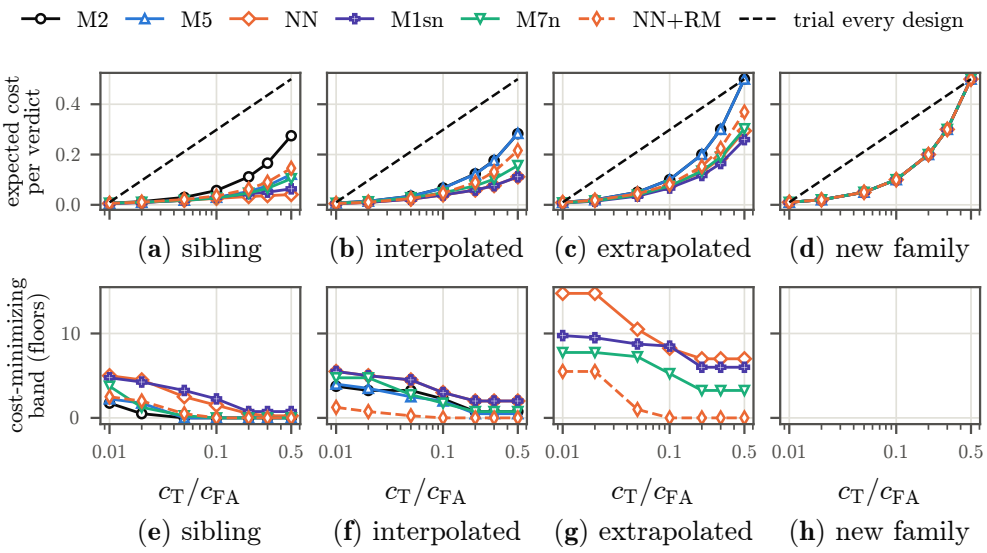


Fig. 6 (a–d) Expected cost per verdict with the cost-minimizing band and (e–h) that band against the cost of a trial (reference prior, $c_{FR} = 0.5\,c_{FA}$, error-free trials, bands of step 5)

trial. The cost-minimizing bands, in sample and pooled over the families, keep the 95 % bands when trials are expensive, and they widen as trials get cheaper; the band of the nearest setting widens to 5–15 floors at a hundredth of a false accept. If trying a new force on an existing tool costs less, as assumed here, the cost-optimal bands are wider. However, a trial, which is priced as error-free in Fig. 6, is not error-free. When scored as a rule, a trial of the first 20 or 50 parts of the run of the design itself decides at 1.35 and 0.90 floors, respectively. The errors of the 20-part trial cost 0.017 of a false accept per verdict (0.037 under the release-like prior), which is as much as the whole expected cost of the nearest setting with a sibling at a trial cost of 0.05 (0.018). Both priors place 47 % of the verdicts on failing designs.

A limit at a performance index of 1.33, which is the lower end of the range in which an initial process study may be acceptable after a customer review, lies a median 1.75 floors above the 95th percentile of an alternative (1.03, 2.46 and 3.14 floors at indices of 1.0, 1.67 and 2.0). Thus, it lies inside every attainable distance except that of the nearest setting when the produced siblings do not differ resolvably (1.60 floors). However, release asks whether the index is at least 1.33, which is a decision on $\bar{m} + 4\sigma$. When scored on this decision, the nearest setting needs 3.35 floors with a sibling and 10.55 floors when extrapolating. Therefore, it confirms an index of 1.33 only for designs whose true within-run index is 2.1–3.3 with a sibling and 3.8–7.4 when extrapolating. Moreover, a short trial does not settle the question once runs differ. With a between-run component of 1, 2 and 3 floors, the 20-part trial decides at 2.1, 3.6 and 5.1 floors, and the nearest setting with a sibling at 3.45, 4.45 and 5.75 floors (`qual_between_run_trial`). Consequently, near release level, the evidence must come from replicated runs.

5.7 Sensitivity, selection and robustness

A between-run component added to every truth lengthens the distances, but it keeps their order. With $\tau = 3$ floors, which is about the largest variation between series, the nearest setting needs 5.8 floors with a sibling and 10 floors when extrapolating, and M5 needs 9.3 and 20 floors. A production-record rule (nearest setting, force trend or median) is the most decisive within the family in each of 200 draws of this component. Selected on one geometry, the most decisive rule is a production-record rule in every relation within a family, and it decides at 2.0–11 floors on the other geometry. For a new family, the rule selected on one transfer decides at 32–33 floors on the other. When selected for safety, the rule chosen on one geometry is safe on the other only at up to 18 floors when interpolating and 11 floors when extrapolating. Furthermore, rule and band selected together keep 0.92–0.97 of the decided verdicts correct within a family (below 0.95 in four of six cases) and 0.73 for a new family (`qual_crossfit`). The paired intervals between point rules span about ± 2 floors with a sibling and when extrapolating, so smaller differences are not resolved by this design.

Requiring all seven characteristics of a design to meet raises the decisive distances of the production-record rules by at most 1.8 floors within a family, but those of M5 by up to 16 floors. Lower limits, truths and calibration from the first 20 or 50 parts, and further variants of data, constants and specifications leave the nearest setting within 1.2 floors of its reference distance and a production-record rule the most decisive within the family. Other floor protocols rescale the distances by 0.60 to 1.42, and floors recomputed without the series of the held-out alternative move the distances of the main rules by at most 1.4 floors and that of M0 by 3.75 floors.

6 Discussion

6.1 Main findings

Two findings follow from the definitions, and they hold wherever the framework is applied. Firstly, the floor puts every rule and characteristic on a common production-referenced scale, though not on a common scale of nonconforming fractions, and as a within-run quantity, it bounds the variation of series production from below (RQ1). Secondly, an interval rule trades decisiveness for safety, which the two distances make explicit. The other findings are observations on one laboratory process, with every alternative produced once and a generic simulation dataset; therefore, they are hypotheses for confirmation. The attainable distance grows with the novelty of the design. A production-record rule applied to every relation decided at 3.4 floors with a sibling and 8.5 floors for a new setting, and no rule decided closer than 30 floors for a new family. Within a family, the choice of rule mattered more than the relation, also among the pre-specified rules (RQ2). For this simulation dataset, and for GPs and boosting fitted to six to nine alternatives, the simulation contributed little within a family, where it helped single characteristics when extrapolating in force; however, it mattered across families for the characteristics that it models directly. The learners added no decisiveness, and what made the rules safe for new settings was the evidence borrowed from the other family on the same relation (RQ3). Finally, the qualification

Table 8 Guidance for design-stage manufacturability decisions and its basis

Situation	Guidance	Basis
From the framework (prescriptive; usefulness not evaluated)		
Before relying on a predictor	Measure the floor on consecutive production with a stated protocol; report σ_{b}, σ_{w}, lag dependence and single-run floors; treat it as a lower bound of series variation.	framework
Qualifying a rule	Evaluate it by grouped cross-validation for the relation of the coming designs, per family and characteristic, siblings stratified by whether they differ.	framework
Choosing a rule	Compare it with the nearest produced setting and with its own version without the simulation, on both distances and by expected cost at the team's trial cost.	framework
Deciding a single design	Take the guard band from the relations the family's production exercises or from another family, validate it out of sample for the expected requirements and widen it where trials are cheap; a margin inside it means a trial.	framework
Observed in this study (one process, 18 alternatives produced once, 2 geometries; not general)		
New variant with a produced sibling	the nearest produced setting decided beyond 3.4 floors (1.6 when the produced siblings did not differ resolvably from each other); no learned rule was more decisive	18 alternatives
New process setting	production record 8.2–8.5 floors, interval rules 15–19; borrowing the other family's errors on the relation made it safe	6 force cells
New family	only rules with the (generic) simulation decided, at 30 floors and beyond; none qualified for single verdicts	two transfers
Release-level requirements	a limit at $P_{\mathrm{pk}} = 1.33$ lies about 2 floors above the 95th percentile; once runs differ by a floor, confirming the index needs replicated runs, not a prediction or a short trial	18 alternatives

by relation exposed a danger: the scaled simulation, which was decisive with a sibling, accepted failing alternatives 165 floors from their limit when extrapolating, and M7 up to 118 floors.

6.2 Implications for practice

Within a family, the production record answers most design-stage questions of a process planner or quality engineer if it is indexed by how the coming design relates to it. A sibling at the same setting settles a requirement a few floors away, and a new force within the produced range does it almost as well; however, an extrapolated force or a new family needs a trial unless the requirement is far. Under the reference prior, of 100 verdicts on new variants with a sibling, the nearest setting with its band sends 2 to trial and gets 92 to 95 of the other 98 right. Of 100 verdicts on extrapolated settings, it sends 58 to trial. Under the release-like prior, it sends 41 and 92 to trial, respectively, and it gets 6 of the other 8 extrapolated verdicts right. Moreover, near release-level limits, once runs differed by a floor, neither a prediction nor a short trial confirmed capability. Therefore, the value of design-stage prediction lies in the decisions far from the limits. For a developer of an AI-based manufacturability tool, the lesson is methodological. A learned model has to be compared with a production-record baseline and with its own version without the simulation, and it has to be scored by evidence relation in a unit that production defines. Its intervals have to be

calibrated on cases of the same relation and shown to its user with a guard band. The nearest setting is also the most transparent rule, and it is the natural default where a learned rule shows no advantage.

6.3 Simulation and machine learning with few produced alternatives

Within a design family, the production record carried the decision even against a friction-fitted (censored) or rescaled simulation. However, the simulation here is a generic dataset that does not contain the produced process (Section 5.2). A process-calibrated simulation, with friction depending on pressure (Hol et al, 2012), a hardening model fit for springback (Wagoner et al, 2013) and elastic tools (Piltham-mar et al, 2018), might contribute more. The per-characteristic results show where the simulation already helps: for the draw-ins and the arm angle when extrapolating, and for the characteristics that it models directly in a new family. The learners failed for a reason. With six to nine alternatives at two or three forces, a flexible model can identify at most a linear trend, and a well-specified one becomes that trend. Nonlinear multi-fidelity models (Perdikaris et al, 2017), physics-informed learning (Karniadakis et al, 2021), conformalized quantile regression on the parts (Romano et al, 2019) and multi-task models over more families (Bonilla et al, 2008) are the next candidates, and they should be scored in the same way.

6.4 Proposed reuse for other processes and AI-based tools

The framework needs several produced variants of a design family whose relations span those of the coming designs, a series of consecutive parts or replicated batches per variant, a design-stage predictor for every variant, and characteristics measured identically on parts and predictions. In this study, nine produced alternatives per family and 500-part series gave ten batches of 50 parts for the floor, one produced sibling sufficed for the nearest setting, and truths from the first 20 or 50 parts moved its distance with a sibling by less than half a floor. The extensions proposed below are untested. A classifier with a fixed threshold does not take the requirement as input. Therefore, its operating characteristic is estimated by binning the test cases by their true margin, which needs a continuous, production-referenced outcome for every test design; CAD-based manufacturability datasets usually lack such outcomes. A language-model check is scored on a grid of requirements with repeated queries. Model, version, date, decoding settings and prompt paraphrases should be reported, refusals and hedged answers should be treated as trials, and the verdicts should be tested for monotonicity in the requirement and for contamination by the public data. A requirement on a rate, such as the share of split parts, is scored once the rate of each alternative is estimated from its series, with a floor over batches of rates. For continuous design spaces, the relation becomes a distance from the produced designs in descriptor space. Table 9 gives a proposed reporting checklist, which is modeled on model cards (Mitchell et al, 2019) and TRIPOD+AI (Collins et al, 2024). The scoring is released as a small documented function that takes intervals, truths and floors and returns distances, guard bands and trial rates (Canseven, 2026). An attempt on

Table 9 Proposed reporting checklist for the evaluation of a design-stage decision rule against production

Item	Why it matters	In this study
Floor and protocol	every distance is in this unit	B, α, lag, single runs (Table 4)
Measurement	biases enter absolute comparisons	Table 3, Section 4.3
Relations, replication	alternatives, not cases, replicate	per relation, 18 alternatives
Both distances	decisiveness and safety differ	Table 5, false-accept side
Baselines	production record, no simulation	NN, M1sn; rules with and without
Intervals	coverage follows the construction	Table 2, Section 5.4
Per characteristic	pooled distances hide the worst	Table 6
Guard band, prior	the deployed part	decision-time, two priors (Table 7)
Costs and trials	deployment is a cost question	Fig. 6
Development record	selection optimism	Section 4.5, cross-fitting
Query protocol	verdicts vary between queries	language models (Section 6.4)

open powder-bed fusion data (Leirmo, 2021) failed, since the data lack ten replicated anchors of one design per build.

6.5 Deployment

In use, a verdict of step 6 would be shown with its margin in floors and in the unit of the characteristic, the relation on which it rests and the produced alternatives behind it. A trial verdict would be shown with the requirement distance at which the rule would decide. A qualification also has to be maintained. The realized errors of the verdicts of step 6 can be tracked against later production on a control chart of decision outcomes; however, such a chart sees false accepts only, since rejected designs are never made. A rule would be re-qualified when its share of wrong decided verdicts exceeds 5 % beyond what the number of verdicts explains (a rise to 10 % takes about 150 decided verdicts to detect at 5 % significance and 80 % power), when the floor of new runs leaves the range of the single-run floors, or when the die is reworked. The evidence has a cost, several 500-part series per family before any rule qualifies, although one produced sibling settles most verdicts with a sibling. Whether teams decide better with the framework is an empirical question. A study with cases of known production outcome could test whether process planners and quality engineers shown the guard-banded verdict accept fewer wrong verdicts within the safe distance than those shown the verdict alone, at an equal rate of right acceptances (Lee and See, 2004). It could also test how often they override a trial verdict and whether the band narrows the spread between teams of the share of wrong verdicts accepted (Zhang et al, 2021).

6.6 Threats to validity

The evaluation covers one material, one press and two geometries, and the relations within a family come from a 3×3 factorial in which lubrication is mostly not resolved. Thus, the numbers are specific to this process, although the procedure is general. The new family rests on two transfers between geometries that differ in three features at once. Each alternative was produced once, so the reference q_p is that of one run, without the variation between days, coils and tool states; the sensitivity analysis explores

its effect, but it cannot replace replicated runs. The floor does not separate scanner drift from process drift, and the absolute values are uncalibrated. The simulation dataset was not calibrated to the process, so the findings on simulation concern this dataset. The requirements are one-sided limits on the 95th percentile of dimensional characteristics, which is below the conformance of series release. In addition, attribute outcomes such as splits and wrinkles are not covered, and the verdicts are marginal per characteristic, a design-level verdict being an extension. The guard band was estimated pooled over the characteristics and, for new settings, borrowed between two families; its prior is a reference rather than the prior of a team, and the costs are assumed. Finally, the rules, strata, bands and priors were partly chosen after the results were seen, and they were evaluated on the same data. The cross-fitting estimates the optimism but does not remove it, and a single analyst carried out the analysis, with code and data open for replication.

7 Conclusion

In this study, a production-referenced evaluation of design-stage decision rules was proposed. The within-run floor, the difference that drift and scatter of production and measurement produce between two batches of one design within a run, was introduced as its unit. The decisive and safe distances in floors were defined to show how close to a requirement a rule decides and how close it can be relied on not to mislead. Moreover, a guard band validated for an assumed prior of requirements was used to turn them into a rule for single decisions, priced by the costs of errors and trials.

The framework was applied, as an exploratory study, to one laboratory deep-drawing process, with alternatives produced once and a generic simulation dataset. It was observed that the attainable distance grew with the novelty of the design. The nearest produced setting, one rule for every relation, decided at 3.4 floors with a produced sibling and 8.5 floors for a new process setting, and no rule decided closer than 30 floors for a new family. Release-level requirements lay within these distances, and once runs differed by a floor, confirming a performance index of 1.33 needed replicated runs, which neither a prediction nor a short trial replaced. It was also seen that the production record carried the decision within a family, whereas across families only the rules using the simulation carried the change of geometry. GPs and boosting fitted to six to nine alternatives reproduced the production record, and the evidence borrowed from the other family on the same relation, not learning, made it safe for new settings.

Based on these findings, it is suggested to index production records by their relation to the coming design rather than to develop a more elaborate model. Within a design family, such a record is the baseline that a learned manufacturability check has to beat. Since the evidence is one process and two geometries, and the comparison of rules is exploratory, future work will focus on four extensions: series from several days, coils and tool states, scanned in a documented order with a reference artifact; a process-calibrated simulation; families with continuous geometric descriptors; and a confirmatory test of the procedure, registered before the analysis, on other data.

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Declarations

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Ethics approval and consent. Not applicable.

Data availability. All data are public: RDDAC (doi:10.18419/DARUS-5589, version 2.0) and DDACS (doi:10.18419/DARUS-4801, version 3.0), both under CC BY 4.0. The extracted feature tables and all result files are archived with the code (Canseven, 2026).

Code availability. The analysis pipeline, which reproduces every number in this paper from the public data, is archived on Zenodo (Canseven, 2026) with the result files named in the text (`results/`); `results/findings.md` maps every number to its script and result file, `results/DATA_DICTIONARY.md` explains the files in the terms of the paper, and `scripts/opchar.py` is the scoring function for other datasets.

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) was used under the author's direction to write and debug code, retrieve and check literature, run analyses and draft and revise text. It was also used to simulate reviews of earlier versions of the manuscript in seven rounds, acting as referees and editor; most rules and analyses added after the first results, listed in Section 4.5, were added in response to these simulated reviews. None is a review by a journal. The author designed the study, verified every analysis against the archived results, checked every reference against its published record, and takes full responsibility for the content; the language model is not an author. All figures come from the released code; none is AI-generated.

References

Arendt PD, Apley DW, Chen W (2012) Quantification of model uncertainty: Calibration, model discrepancy, and identifiability. Journal of Mechanical Design 134(10):100908. https://doi.org/10.1115/1.4007390

ASME (2018) Assessing credibility of computational modeling through verification and validation: Application to medical devices. ASME V&V 40-2018, American Society of Mechanical Engineers, New York

Automotive Industry Action Group (2006) Production Part Approval Process (PPAP), 4th edn. AIAG, Southfield, MI

Automotive Industry Action Group (2010) Measurement Systems Analysis, 4th edn. AIAG, Southfield, MI

Ballegeer M, Benoit DF (2026) BenDFM: A taxonomy and synthetic CAD dataset for manufacturability assessment in sheet metal bending. Journal of Intelligent Manufacturing https://doi.org/10.1007/s10845-026-02857-9

Banabic D (2010) Sheet Metal Forming Processes: Constitutive Modelling and Numerical Simulation. Springer, Berlin, Heidelberg, https://doi.org/10.1007/978-3-540-88113-1

Barber RF, Candès EJ, Ramdas A, et al (2021) Predictive inference with the jackknife+. The Annals of Statistics 49(1):486–507. https://doi.org/10.1214/20-aos1965

Barber RF, Candès EJ, Ramdas A, et al (2023) Conformal prediction beyond exchangeability. The Annals of Statistics 51(2):816–845. https://doi.org/10.1214/23-aos2276

Bates S, Hastie T, Tibshirani R (2024) Cross-validation: What does it estimate and how well does it do it? Journal of the American Statistical Association 119(546):1434–1445. https://doi.org/10.1080/01621459.2023.2197686

Baum S, Heinzelmann P (2025) Deep drawing and cutting simulations (DDACS) dataset. DaRUS, University of Stuttgart, version 3.0, https://doi.org/10.18419/DARUS-4801

Baum S, Heinzelmann P (2026) Real deep drawing and cutting (RDDAC) dataset. DaRUS, University of Stuttgart, version 2.0, https://doi.org/10.18419/DARUS-5589

Baum S, Heinzelmann P, Clauß P, et al (2026a) Statistical analysis of simulation-to-reality deviation in deep drawing with a benchmark dataset. Transactions of the Indian Institute of Metals 79(7):176. https://doi.org/10.1007/s12666-026-03870-5

Baum S, Heinzelmann P, Liewald M, et al (2026b) Addressing discrepancies between synthetic and real-world data of deep-drawing manufacturing systems in AI applications. Procedia CIRP 138:727–732. https://doi.org/10.1016/j.procir.2026.01.125

Bechhofer RE (1954) A single-sample multiple decision procedure for ranking means of normal populations with known variances. The Annals of Mathematical Statistics 25(1):16–39. https://doi.org/10.1214/aoms/1177728845

Bonilla EV, Chai KMA, Williams CKI (2008) Multi-task Gaussian process prediction. In: Advances in Neural Information Processing Systems, vol 20. Curran Associates, Inc., pp 153–160

Boothroyd G, Dewhurst P, Knight WA (2010) Product Design for Manufacture and Assembly, 3rd edn. CRC Press, Boca Raton, https://doi.org/10.1201/9781420089288

Brynjarsdóttir J, O'Hagan A (2014) Learning about physical parameters: the importance of model discrepancy. Inverse Problems 30(11):114007. https://doi.org/10.1088/0266-5611/30/11/114007

Canseven HT (2026) Code, feature tables and results for "A production-referenced evaluation of design-stage decision rules in deep drawing: when can a prediction replace a physical trial?". Zenodo, version 3, *[DOI pending]*

Cao J, Bambach M, Merklein M, et al (2024) Artificial intelligence in metal forming. CIRP Annals 73(2):561–587. https://doi.org/10.1016/j.cirp.2024.04.102

Cawley GC, Talbot NLC (2010) On over-fitting in model selection and subsequent selection bias in performance evaluation. Journal of Machine Learning Research 11:2079–2107. URL https://jmlr.org/papers/v11/cawley10a.html

Chen K, Breunig A, Ha J, et al (2022) Robustness of deep-drawing finite-element simulations to process variations. International Journal of Material Forming 15(3):45. https://doi.org/10.1007/s12289-022-01695-3

Cheng FT, Chen YT, Su YC, et al (2008) Evaluating reliance level of a virtual metrology system. IEEE Transactions on Semiconductor Manufacturing 21(1):92–103. https://doi.org/10.1109/tsm.2007.914373

Cheng FT, Huang HC, Kao CA (2012) Developing an automatic virtual metrology system. IEEE Transactions on Automation Science and Engineering 9(1):181–188. https://doi.org/10.1109/tase.2011.2169405

Chow CK (1970) On optimum recognition error and reject tradeoff. IEEE Transactions on Information Theory 16(1):41–46. https://doi.org/10.1109/TIT.1970.1054406

Collins GS, Moons KGM, Dhiman P, et al (2024) TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ 385:e078378. https://doi.org/10.1136/bmj-2023-078378

Dreyfus PA, Psarommatis F, May G, et al (2022) Virtual metrology as an approach for product quality estimation in Industry 4.0: a systematic review and integrative conceptual framework. International Journal of Production Research 60(2):742–765. https://doi.org/10.1080/00207543.2021.1976433

Eckert C, Isaksson O, Earl C (2019) Design margins: a hidden issue in industry. Design Science 5:e9. https://doi.org/10.1017/dsj.2019.7

Elkan C (2001) The foundations of cost-sensitive learning. In: Proceedings of the Seventeenth International Joint Conference on Artificial Intelligence (IJCAI 2001). Morgan Kaufmann, pp 973–978

Fischer P, Harsch D, Heingärtner J, et al (2018) Implementation of feedback control in kitchen sink production. IOP Conference Series: Materials Science and Engineering 418:012110. https://doi.org/10.1088/1757-899x/418/1/012110
Friedman JH (2001) Greedy function approximation: A gradient boosting machine. The Annals of Statistics 29(5):1189–1232. https://doi.org/10.1214/aos/1013203451
Geifman Y, El-Yaniv R (2017) Selective classification for deep neural networks. In: Advances in Neural Information Processing Systems, vol 30. Curran Associates, Inc., pp 4878–4887
Ghadai S, Balu A, Sarkar S, et al (2018) Learning localized features in 3D CAD models for manufacturability analysis of drilled holes. Computer Aided Geometric Design 62:263–275. https://doi.org/10.1016/j.cagd.2018.03.024
Gneiting T, Raftery AE (2007) Strictly proper scoring rules, prediction, and estimation. Journal of the American Statistical Association 102(477):359–378. https://doi.org/10.1198/016214506000001437
Havinga J, Mandal PK, van den Boogaard T (2020) Exploiting data in smart factories: real-time state estimation and model improvement in metal forming mass production. International Journal of Material Forming 13(5):663–673. https://doi.org/10.1007/s12289-019-01495-2
Hazelrigg GA (1998) A framework for decision-based engineering design. Journal of Mechanical Design 120(4):653–658. https://doi.org/10.1115/1.2829328
Heddoub A, Diallo AR, Homri L, et al (2026) Conformal prediction for multivariate quality assurance – application on coordinate measuring machines. Procedia CIRP 145:91–96. https://doi.org/10.1016/j.procir.2026.03.151
Heinzelmann P, Baum S, Riedmüller KR, et al (2025a) A comprehensive benchmark dataset for sheet metal forming: Advancing machine learning and surrogate modelling in process simulations. MATEC Web of Conferences 408:01090. https://doi.org/10.1051/matecconf/202540801090
Heinzelmann P, Baum S, Riedmüller KR, et al (2025b) A data-driven surrogate model for predicting springback in deep drawing processes. Journal of Physics: Conference Series 3104(1):012060. https://doi.org/10.1088/1742-6596/3104/1/012060
Higdon D, Kennedy M, Cavendish JC, et al (2004) Combining field data and computer simulations for calibration and prediction. SIAM Journal on Scientific Computing 26(2):448–466. https://doi.org/10.1137/s1064827503426693
Hol J, Cid Alfaro MV, de Rooij MB, et al (2012) Advanced friction modeling for sheet metal forming. Wear 286–287:66–78. https://doi.org/10.1016/j.wear.2011.04.004
ISO (1994) Accuracy (trueness and precision) of measurement methods and results – part 6: Use in practice of accuracy values. ISO 5725-6:1994, International Organization for Standardization, Geneva
ISO (2017) Geometrical product specifications (GPS) – inspection by measurement of workpieces and measuring equipment – part 1: Decision rules for verifying conformity or nonconformity with specifications. ISO 14253-1:2017, International Organization for Standardization, Geneva
ISO (2023) Accuracy (trueness and precision) of measurement methods and results – part 3: Intermediate precision and alternative designs for collaborative studies. ISO 5725-3:2023, International Organization for Standardization, Geneva
ISO (2025) Accuracy (trueness and precision) of measurement methods and results – part 2: Basic method for the determination of repeatability and reproducibility of a standard measurement method. ISO 5725-2:2025, International Organization for Standardization, Geneva
ISO (2026) Statistical methods in process management – capability and performance – part 2: Process capability and performance of time-dependent process models. ISO 22514-2:2026, International Organization for Standardization, Geneva
Joint Committee for Guides in Metrology (2012) Evaluation of measurement data – the role of measurement uncertainty in conformity assessment. JCGM 106:2012, BIPM, Sèvres, https://doi.org/10.59161/jcgm106-2012
Karniadakis GE, Kevrekidis IG, Lu L, et al (2021) Physics-informed machine learning. Nature Reviews Physics 3(6):422–440. https://doi.org/10.1038/s42254-021-00314-5
Kennedy MC, O'Hagan A (2000) Predicting the output from a complex computer code when fast approximations are available. Biometrika 87(1):1–13. https://doi.org/10.1093/biomet/87.1.1
Kennedy MC, O'Hagan A (2001) Bayesian calibration of computer models. Journal of the Royal Statistical Society Series B: Statistical Methodology 63(3):425–464. https://doi.org/10.1111/1467-9868.00294

Le Gratiet L, Garnier J (2014) Recursive co-kriging model for design of computer experiments with multiple levels of fidelity. International Journal for Uncertainty Quantification 4(5):365–386. https://doi.org/10.1615/int.j.uncertaintyquantification.2014006914

Lee JD, See KA (2004) Trust in automation: Designing for appropriate reliance. Human Factors: The Journal of the Human Factors and Ergonomics Society 46(1):50–80. https://doi.org/10.1518/hfes.46.1.50.30392

Lei J, G'Sell M, Rinaldo A, et al (2018) Distribution-free predictive inference for regression. Journal of the American Statistical Association 113(523):1094–1111. https://doi.org/10.1080/01621459.2017.1307116

Leirmo TL (2021) Data on dimensional and geometric accuracy for laser-based powder bed fusion of PA12 (PA2200). DataverseNO, version 1.1, https://doi.org/10.18710/DHACHZ

Madras D, Pitassi T, Zemel R (2018) Predict responsibly: Improving fairness and accuracy by learning to defer. In: Advances in Neural Information Processing Systems, vol 31. Curran Associates, Inc., pp 6147–6157

Makatura L, Foshey M, Wang B, et al (2024) How can large language models help humans in design and manufacturing? Part 1: Elements of the LLM-enabled computational design and manufacturing pipeline. Harvard Data Science Review (Special Issue 5). https://doi.org/10.1162/99608f92.cc80fe30

Maschler B, Weyrich M (2021) Deep transfer learning for industrial automation: A review and discussion of new techniques for data-driven machine learning. IEEE Industrial Electronics Magazine 15(2):65–75. https://doi.org/10.1109/mie.2020.3034884

Mitchell M, Wu S, Zaldivar A, et al (2019) Model cards for model reporting. In: Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19). Association for Computing Machinery, pp 220–229, https://doi.org/10.1145/3287560.3287596

Montgomery DC (2019) Introduction to Statistical Quality Control, 8th edn. Wiley, Hoboken, NJ

Nemani V, Biggio L, Huan X, et al (2023) Uncertainty quantification in machine learning for engineering design and health prognostics: A tutorial. Mechanical Systems and Signal Processing 205:110796. https://doi.org/10.1016/j.ymssp.2023.110796

Oberkampf WL, Roy CJ (2010) Verification and Validation in Scientific Computing. Cambridge University Press, Cambridge, https://doi.org/10.1017/cbo9780511760396

Pahl G, Beitz W, Feldhusen J, et al (2007) Engineering Design: A Systematic Approach, 3rd edn. Springer, London, https://doi.org/10.1007/978-1-84628-319-2

Pedregosa F, Varoquaux G, Gramfort A, et al (2011) Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12(85):2825–2830. URL https://jmlr.org/papers/v12/pedregosa11a.html

Perdikaris P, Raissi M, Damianou A, et al (2017) Nonlinear information fusion algorithms for data-efficient multi-fidelity modelling. Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences 473(2198):20160751. https://doi.org/10.1098/rspa.2016.0751

Picard C, Edwards KM, Doris AC, et al (2025) From concept to manufacturing: evaluating vision-language models for engineering design. Artificial Intelligence Review 58(9):288. https://doi.org/10.1007/s10462-025-11290-y

Pilthammar J, Sigvant M, Kao-Walter S (2018) Introduction of elastic die deformations in sheet metal forming simulations. International Journal of Solids and Structures 151:76–90. https://doi.org/10.1016/j.ijsolstr.2017.05.009

Rasmussen CE, Williams CKI (2006) Gaussian Processes for Machine Learning. MIT Press, Cambridge, MA, https://doi.org/10.7551/mitpress/3206.001.0001

Roberts DR, Bahn V, Ciuti S, et al (2017) Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40(8):913–929. https://doi.org/10.1111/ecog.02881

Romano Y, Patterson E, Candès EJ (2019) Conformalized quantile regression. In: Advances in Neural Information Processing Systems, vol 32. Curran Associates, Inc., pp 3538–3548

Salado A, Kannan H (2018) A mathematical model of verification strategies. Systems Engineering 21(6):593–608. https://doi.org/10.1002/sys.21463

Tercan H, Meisen T (2022) Machine learning and deep learning based predictive quality in manufacturing: a systematic review. Journal of Intelligent Manufacturing 33(7):1879–1905. https://doi.org/10.1007/s10845-022-01963-8

Tercan H, Guajardo A, Heinisch J, et al (2018) Transfer-learning: Bridging the gap between real and simulation data for machine learning in injection molding. Procedia CIRP 72:185–190. https://doi.org/10.1016/j.procir.2018.03.087

Thomke S, Bell DE (2001) Sequential testing in product development. Management Science 47(2):308–323. https://doi.org/10.1287/mnsc.47.2.308.9838

Varma S, Simon R (2006) Bias in error estimation when using cross-validation for model selection. BMC Bioinformatics 7:91. https://doi.org/10.1186/1471-2105-7-91

Vickers AJ, Elkin EB (2006) Decision curve analysis: A novel method for evaluating prediction models. Medical Decision Making 26(6):565–574. https://doi.org/10.1177/0272989x06295361

Vovk V, Gammerman A, Shafer G (2005) Algorithmic Learning in a Random World. Springer, New York, https://doi.org/10.1007/b106715

Wagoner RH, Lim H, Lee MG (2013) Advanced issues in springback. International Journal of Plasticity 45:3–20. https://doi.org/10.1016/j.ijplas.2012.08.006

Whitney DE (1996) Why mechanical design cannot be like VLSI design. Research in Engineering Design 8(3):125–138. https://doi.org/10.1007/bf01608348

Wiebenga JH, Atzema EH, An YG, et al (2014) Effect of material scatter on the plastic behavior and stretchability in sheet metal forming. Journal of Materials Processing Technology 214(2):238–252. https://doi.org/10.1016/j.jmatprotec.2013.08.008

Wuest T, Weimer D, Irgens C, et al (2016) Machine learning in manufacturing: advantages, challenges, and applications. Production & Manufacturing Research 4(1):23–45. https://doi.org/10.1080/21693277.2016.1192517

Yan X, Wang Z, Zhao C, et al (2026) Machine learning-based cyber manufacturing services: A review of manufacturing process selection, process planning, and design for manufacturing. Engineering 63:47–67. https://doi.org/10.1016/j.eng.2025.11.034

Yoshida F, Uemori T (2002) A model of large-strain cyclic plasticity describing the Bauschinger effect and workhardening stagnation. International Journal of Plasticity 18(5–6):661–686. https://doi.org/10.1016/S0749-6419(01)00050-X

Zaoui A, Denis C, Hebiri M (2020) Regression with reject option and application to kNN. In: Advances in Neural Information Processing Systems, vol 33. Curran Associates, Inc., pp 20073–20082

Zhang G, Raina A, Cagan J, et al (2021) A cautionary tale about the impact of AI on human design teams. Design Studies 72:100990. https://doi.org/10.1016/j.destud.2021.100990

Zhang Y, Liao QV, Bellamy RKE (2020) Effect of confidence and explanation on accuracy and trust calibration in AI-assisted decision making. In: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT* '20). Association for Computing Machinery, pp 295–305, https://doi.org/10.1145/3351095.3372852

Zhang Z, Jaiswal P, Rai R (2018) FeatureNet: Machining feature recognition based on 3D convolution neural network. Computer-Aided Design 101:12–22. https://doi.org/10.1016/j.cad.2018.03.006